

Bill of Materials

Urine Analyzer Lite | India Sourcing | Prices in INR (approx. ₹84 = \$1 USD)

Ref	Component	Description	Qty	Unit (₹)	Total (₹)	India Supplier	Sourcing Note
U1	ESP32-S3-DevKitC-1	Dual-core MCU with native USB; 8MB flash + 8MB OPI PSRAM	1	1299	1299	Robocraze	Also on Amazon.in ~₹1350; Evelta stocks too
U2	TP4056 USB-C Module	LiPo charging module with USB-C input and protection circuit	1	65	65	Robocraze	Very common; also at Probots / Amazon.in ~₹60
U3	S13V25F5 Buck-Boost Regulator	5 V 2.5 A step-up/step-down regulator from LiPo	1	1099	1099	Mouser India	Pololu product; Mouser.in stocks it. element14.in alternate
BT1	3.7V 2000mAh LiPo	Main rechargeable power source	1	599	599	Robocraze	Probots also carries; verify protection circuit included
U4	TCS34725 RGB Sensor #1	I2C RGB+clear color sensor with IR cut filter (primary pad sensor)	1	649	649	Robocraze	Evelta ~₹700; confirm IR-cut filter present on module
U5	TCS34725 RGB Sensor #2	I2C RGB+clear color sensor (secondary / differential)	1	649	649	Robocraze	Same part as U4; buy 2 together for small discount
U6	SHT31-D Module	High-accuracy I2C temperature and humidity sensor	1	749	749	Probots	Robocraze alternate ~₹780; avoid clone modules for accuracy
U7	BH1750 FVI	I2C digital ambient light sensor	1	149	149	Robocraze	Amazon.in ~₹160; very common; GY-302 breakout widely available
U8	ADS1115 ADC Module	16-bit I2C ADC with PGA for NTC thermistor	1	449	449	Robocraze	Probots ~₹400; Evelta ~₹480
RT1	NTC Thermistor 10k B3950	Temperature sensing inside optical chamber	1	25	25	Electronicscomp	Sunrom.com ~₹20; sold in packs of 5 — buy pack
U9	MX1508 Dual H-Bridge	Dual DC motor driver 2A per channel	1	120	120	Robocraze	Amazon.in ~₹100; Robokits ~₹110
M1	GA12-N20 6V 200RPM + Encoder	Compact DC gear motor with Hall-effect quadrature encoder	1	449	449	Robocraze	Robokits ~₹500; confirm 6V 200RPM variant with encoder
SW1	KW12-3 Micro Limit Switch	Home position limit switch (NO)	1	65	65	Robocraze	Amazon.in ₹50–80; buy pack of 5 for spares
SW2	KW12-3 Micro Limit Switch	End-of-travel limit switch (NO)	1	65	65	Robocraze	Same as SW1; order 4 total (2 working + 2 spares)
D1	High-CRI 5mm LED 95+ CRI	High colour-rendering white illumination LED for pad reading	1	220	220	Mouser India	IMPORT — no Indian-stocked 95+CRI 5mm LED equivalent; Nichia NSPWR70CSS-K1 via Mouser.in. Budget alternative: 90CRI Cree-equiv from Amazon.in ~₹80
Q1	2N7000 N-Channel MOSFET	PWM control MOSFET for illumination LED	1	15	15	Electronicscomp	Sunrom.com ~₹20; sold loose — buy 10 for <₹100
D2	275nm UVC LED Module	UV-C sterilization LED for tray decontamination	1	1050	1050	Mouser India	IMPORT — Mouser.in stocks Klaran / SETi UVC LEDs; order with heatsink. element14.in alternate. Allow 2-week lead time
U10	HX711 Load Cell Amplifier Kit 1kg	24-bit ADC for load cell weight measurement	1	320	320	Robocraze	Includes 1 kg bar-type load cell; Probots ~₹350

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PR1	CSN-A2 Micro Thermal Printer	58mm thermal receipt printer for diagnostic reports	1	2800	2800	Robocraze	Amazon.in ■2500–3200 depending on seller; verify TTL UART version (not USB-only)
LCD1	Waveshare 4.3in DSI LCD	Capacitive touch display; MIPI DSI interface	1	3999	3999	Evelta	Robocraze ~■4200; verify DSI cable included. Waveshare India authorised reseller: robu.in
SW3	Momentary Tactile Button 6x6mm	Scan trigger button	1	5	5	Electronicscomp	Sunrom ~■4; buy strip of 20 (<■80 total)
--	GT2 Timing Belt 200mm 6mm wide	Belt drive for strip-tray linear motion	1	199	199	Robocraze	Amazon.in ~■180; buy 2 (one spare)
--	GT2 Pulley 16T 3mm bore	Aluminium motor and idler pulleys	2	175	350	Robocraze	Amazon.in ~■150 each
--	M3 Stainless Socket Head Screws M3×10	General chassis fasteners	20	4	80	Amazon India	Buy 50-pack ~■150 for spares
--	M3 Brass Heat-Set Inserts M3×5	Threaded inserts for 3D-printed parts	12	8	96	Amazon India	Robocraze carries too; buy 20-pack
--	Frosted Acrylic Sheet 50×20×2mm	LED diffuser plate in optical chamber	1	180	180	Amazon India	Cut from A4 sheet ~■300; local acrylic shop in most cities cheaper
--	2mm Clear Acrylic Sheet (laser cut)	Strip flattening guard over tray	1	250	250	Amazon India	Local laser cutting shop (Lamington Rd Mumbai / Nehru Place Delhi) cheaper than online
--	Aluminium Sheet 80×40×2mm	UV-C safety shield (internal)	1	320	320	Amazon India	Local hardware/metal supplier; cut from 200×200mm sheet
--	Spectralon 99% White Standard 10×10mm	White calibration standard for sensor normalisation	1	2200	2200	Labsphere (import)	IMPORT — no India distributor; ship via FedEx International. DIY alternative: pressed BaSO4 tablet (~■200 at lab chemicals supplier like HiMedia)
--	Whatman Grade 1 Filter Paper Strips	Disposable tray liners pack of 100	100	12	1200	Merck India	HiMedia Laboratories (Mumbai/Delhi/Chennai) alternate; also Amazon.in ~■1400 for pack
--	Chassis Base (3D print – PLA)	Compact main chassis 160×120×55mm	1	300	300	Self / Local bureau	PLA ~■900/kg on Amazon.in (eSun / Bambu); 3D printing bureaus in major cities ~■5–8/gram
--	Optical Chamber (3D print – Black PLA)	Light-tight sensor housing 65×30×45mm	1	175	175	Self / Local bureau	Must be Black PLA; 100% infill adds material cost
--	Dipstick Tray (3D print – PLA)	Single 6mm strip channel with filter-paper slot	1	175	175	Self / Local bureau	Print in natural/white PLA; replace each batch run if contaminated
--	MCU Tray (3D print – PLA)	Snap-fit bracket for ESP32-S3-DevKitC-1	1	45	45	Self / Local bureau	Simple flat part; low print cost
--	Printer Bracket (3D print – PETG)	Mount for CSN-A2 thermal printer	1	220	220	Self / Local bureau	PETG ~■1200/kg; use for heat-resistant parts near printer
--	Display Bezel (3D print – PLA)	Front panel with 4.3in LCD cutout	1	130	130	Self / Local bureau	Sand and paint for professional finish
--	UV-C Shield (3D print – Black PLA)	Internal shroud for UV-C LED	1	90	90	Self / Local bureau	100% infill; UV-stable PLA preferred
--	Belt Tensioner (3D print – PLA)	Adjustable GT2 idler arm	1	90	90	Self / Local bureau	Pair with 625ZZ bearing (■60 on Amazon.in)
--	Limit Switch Bracket ×2 (3D print – PLA)	Adjustable KW12-3 mount	2	25	50	Self / Local bureau	Small part; minimal material

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Ref	Component	Description	Qty	Unit (■)	Total (■)	India Supplier	Sourcing Note
--	N20 Motor Bracket (3D print – PLA)	Adapter to mount N20 motor to chassis	1	45	45	Self / Local bureau	Print in PETG if near motor heat
				TOTAL (INR)		26156	
				TOTAL (USD approx @ ■84)		~311	

Note: Items marked IMPORT require ordering via Mouser India (mouser.in) or element14 India (in.element14.com). Allow 1–2 week lead time. 3D-printed parts use self-sourced PLA/PETG filament (~■900–1200/kg on Amazon.in).